

FraudShield AI

Adaptive Fraud Detection with Explainable Risk Intelligence

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FrostHack Financial Services Track

System Architecture

FraudShield AI is a multi-layered fraud detection system featuring 18 detection layers, a 4-model stacking ensemble with Logistic Regression meta-learner, temporal graph analysis using NetworkX and Louvain community detection, SHAP-based explainability, Optuna hyperparameter tuning, target encoding, and adversarial robustness testing.

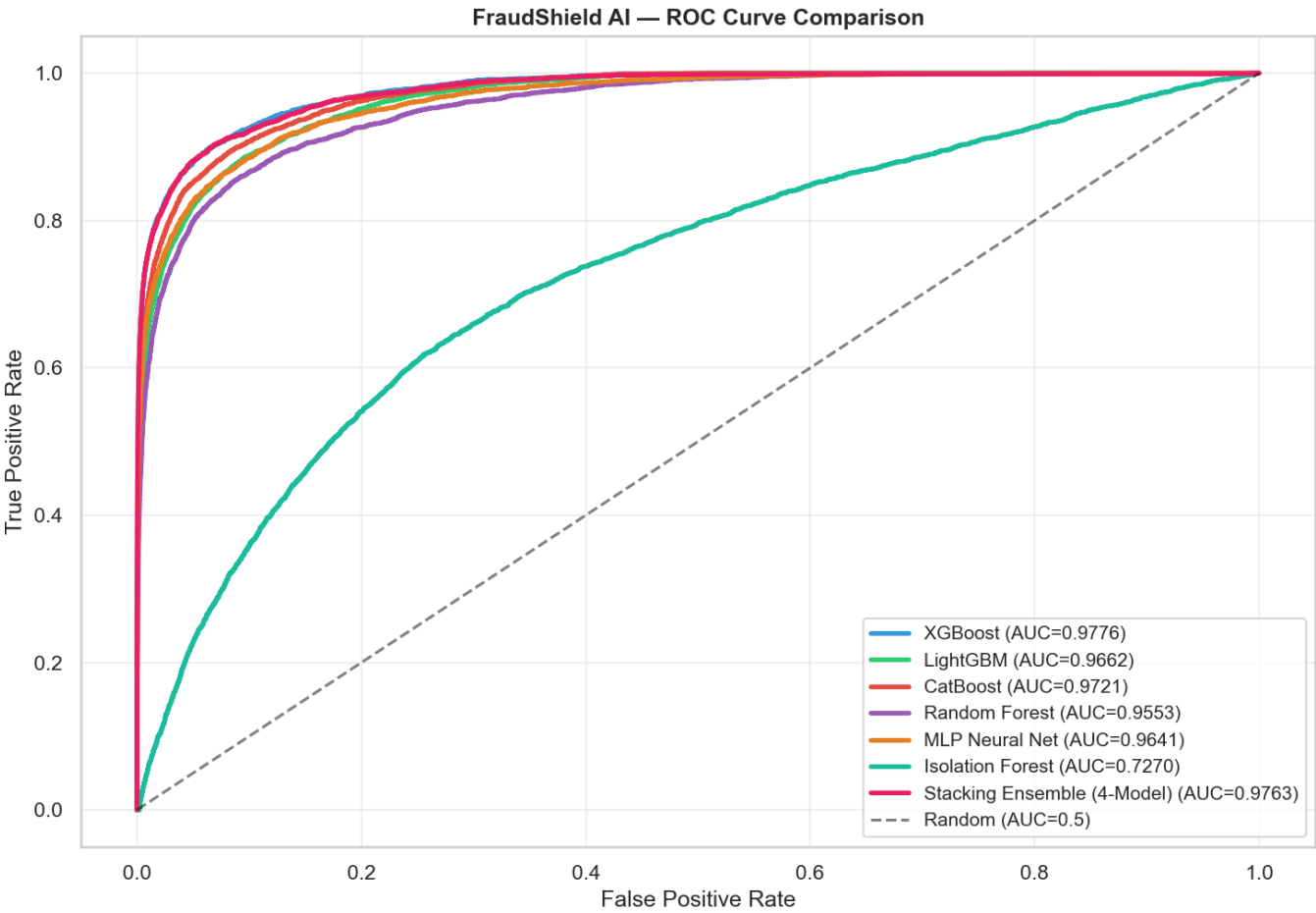
The system includes a FastAPI real-time scoring API, a 7-page Streamlit dashboard, and an ROI cost-benefit calculator for business impact analysis.

Model Performance

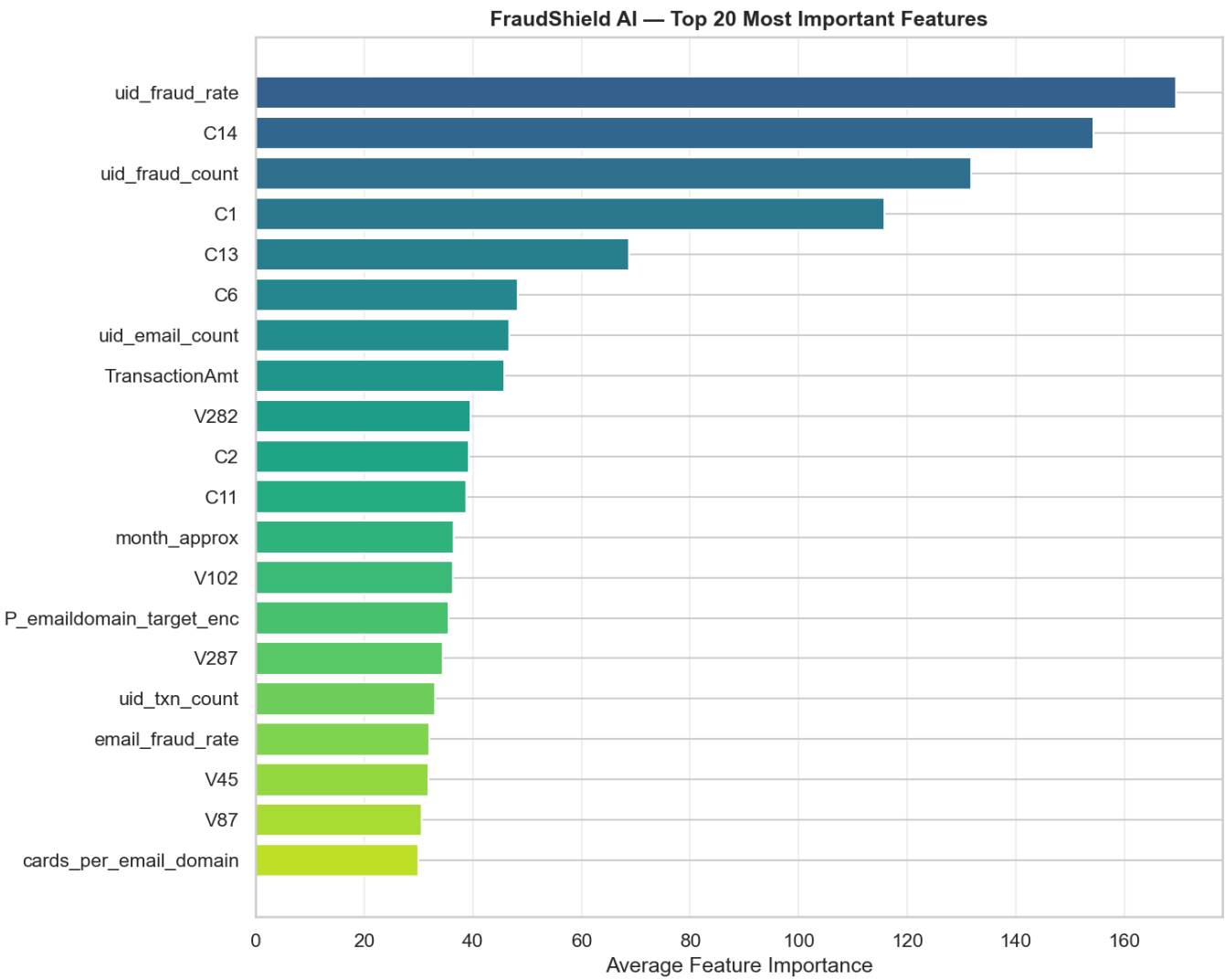
Model	AUC	F1	Precision	Recall
XGBoost	0.9776	0.7474	0.8996	0.6392
LightGBM	0.9662	0.5915	0.4837	0.7612
CatBoost	0.9721	0.6592	0.5806	0.7624
Random Forest	0.9553	0.5614	0.4507	0.7443
MLP Neural Net	0.9641	0.6944	0.8239	0.6000
Isolation Forest	0.7270	0.0948	0.1697	0.0658
Stacking Ensemble (4	0.9763	0.7651	0.8337	0.7070

Best Individual Model: XGBoost (AUC: 0.9776)

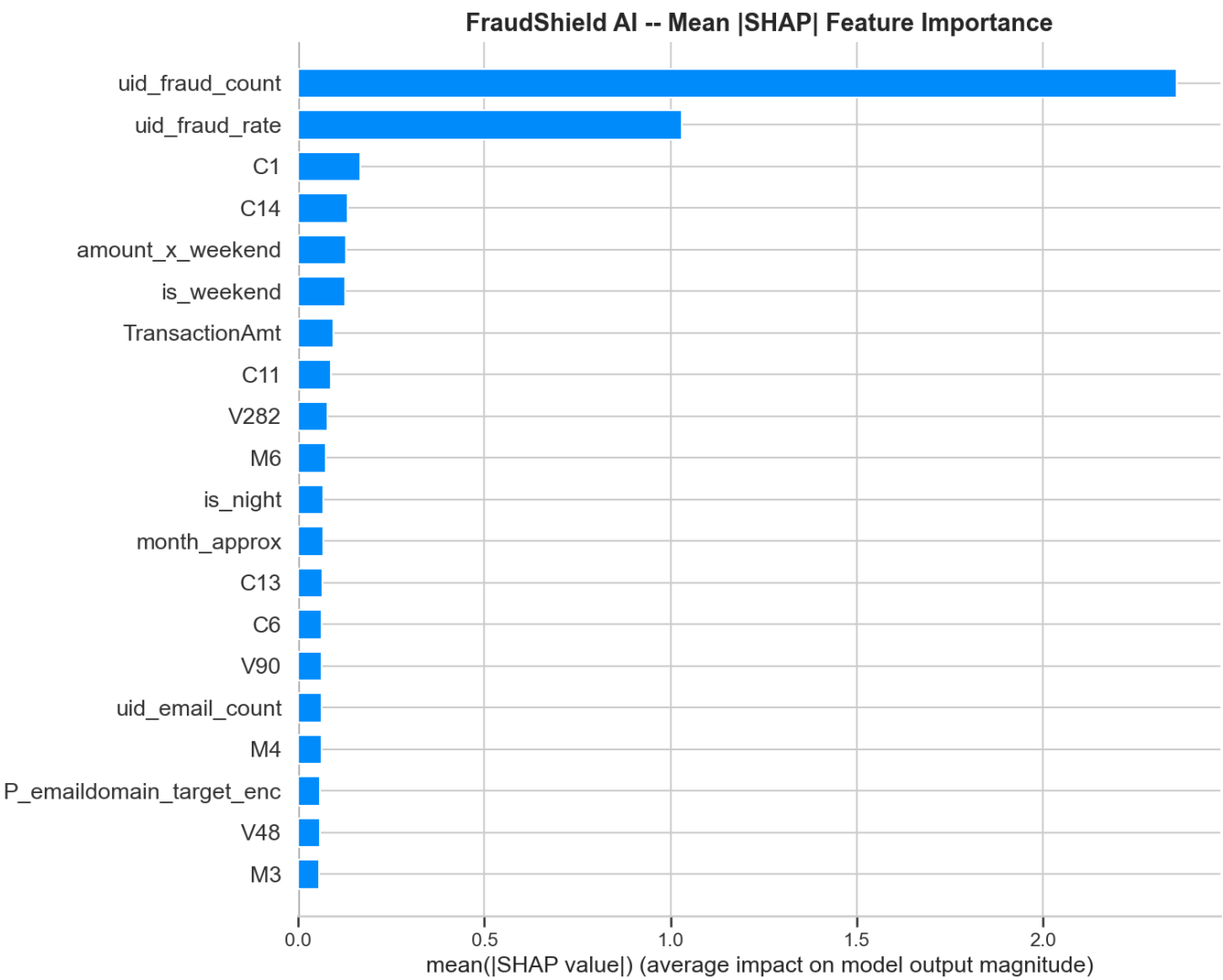
Stacking Ensemble: AUC 0.9763 | F1 0.7651 | Precision 0.8337



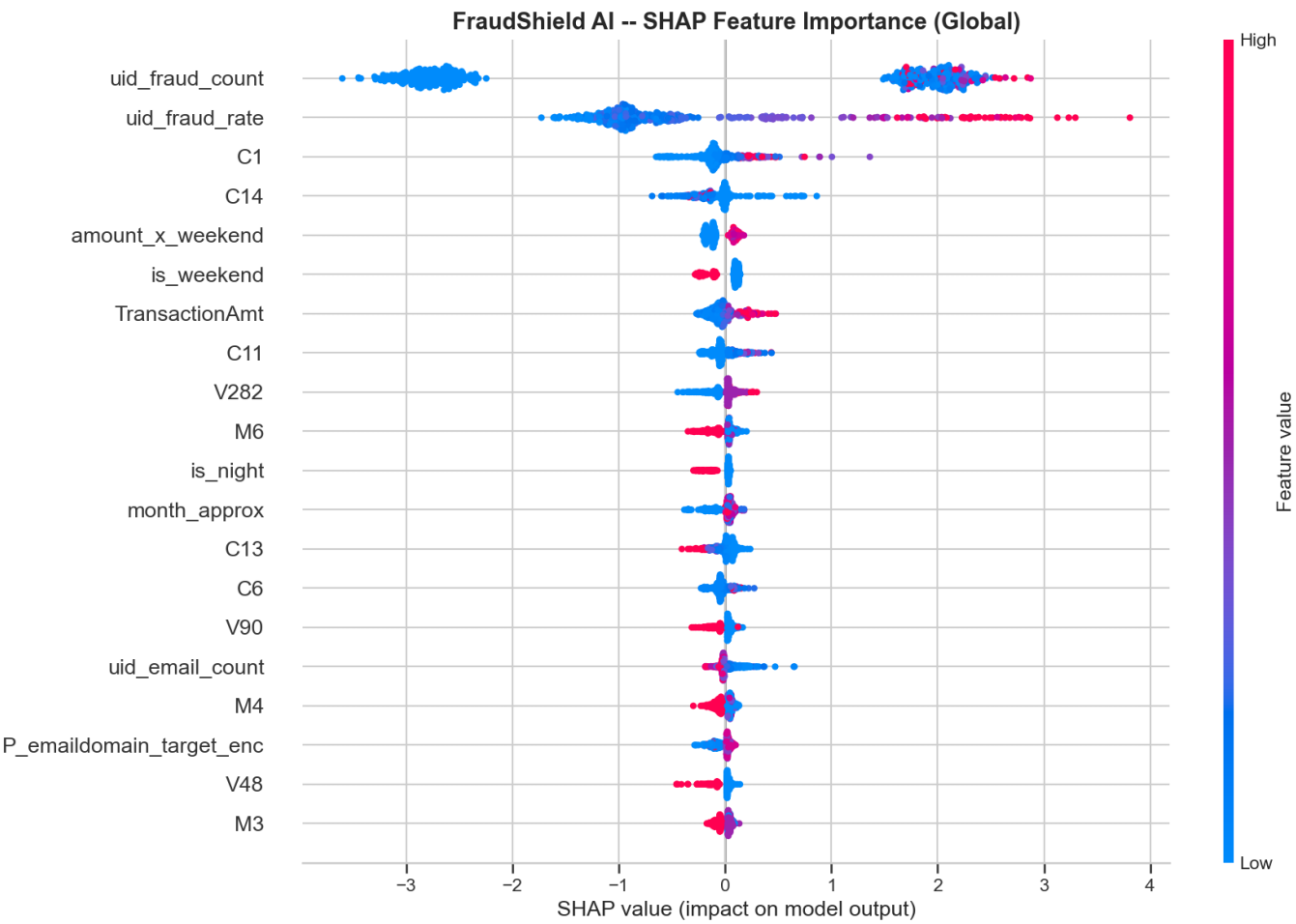
Feature Importance & Explainability



SHAP Feature Importance (Mean |SHAP|)

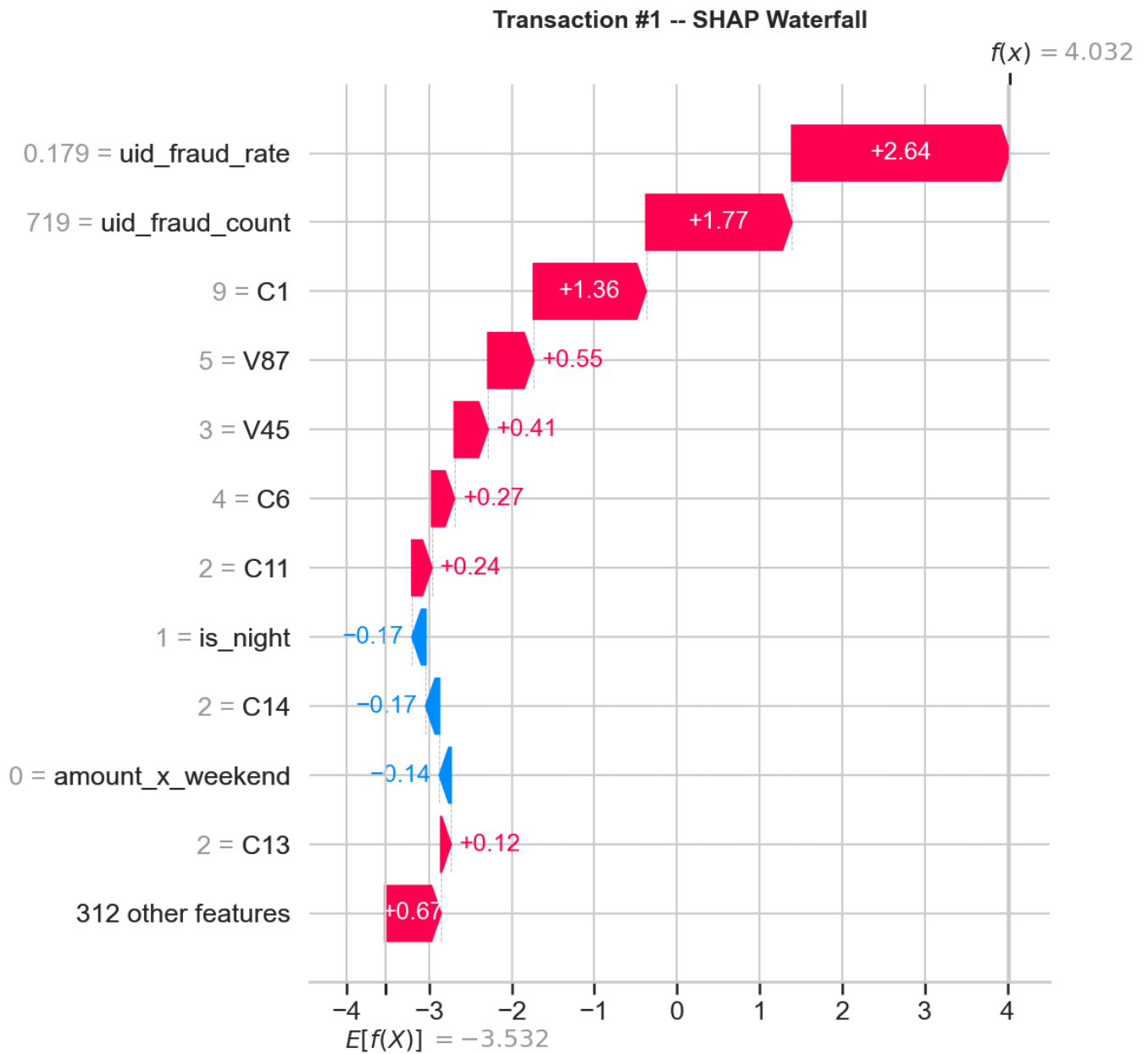


SHAP Beeswarm Summary



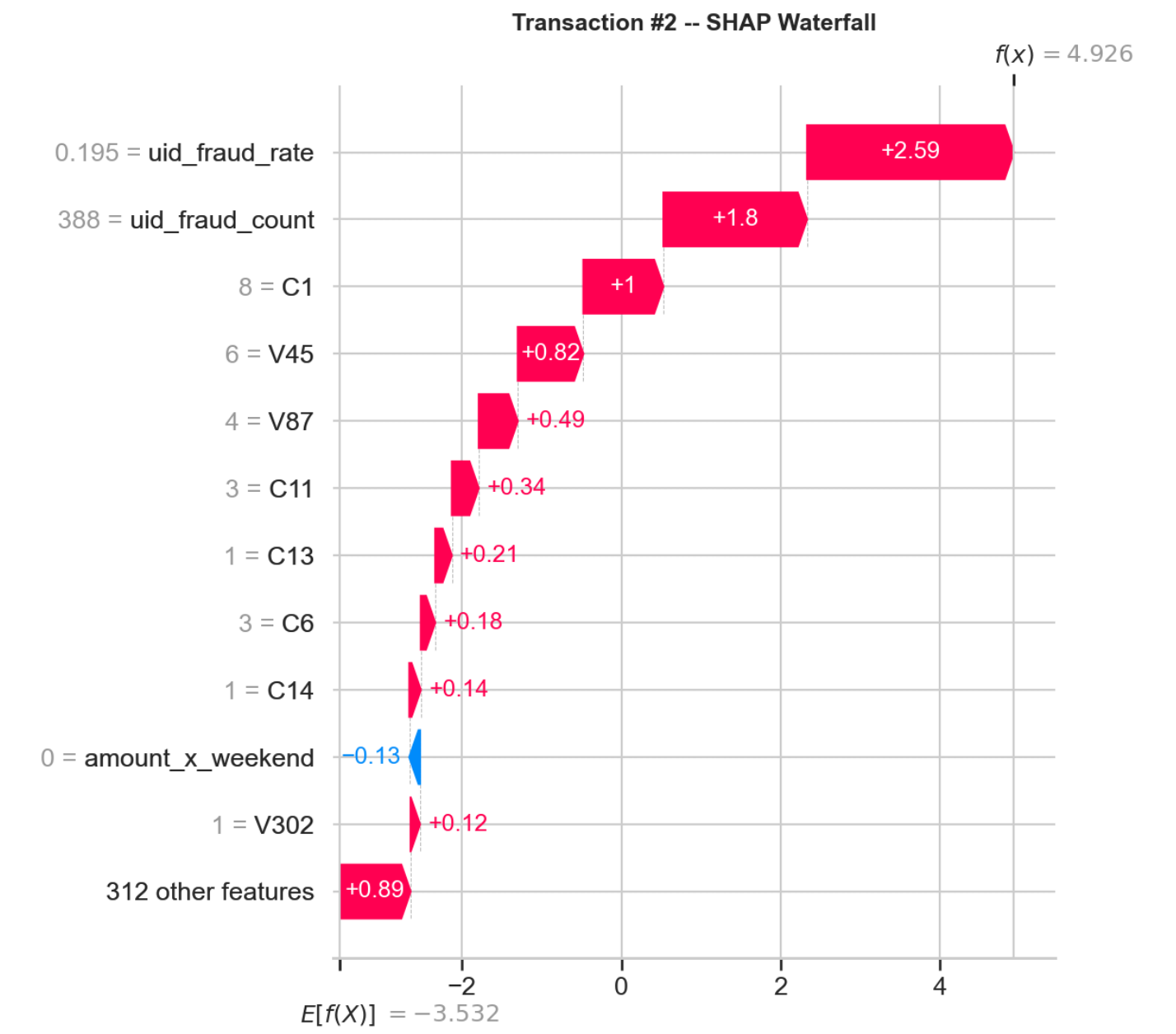
SHAP Waterfall -- Transaction #1

Individual feature contributions for a high-risk transaction:



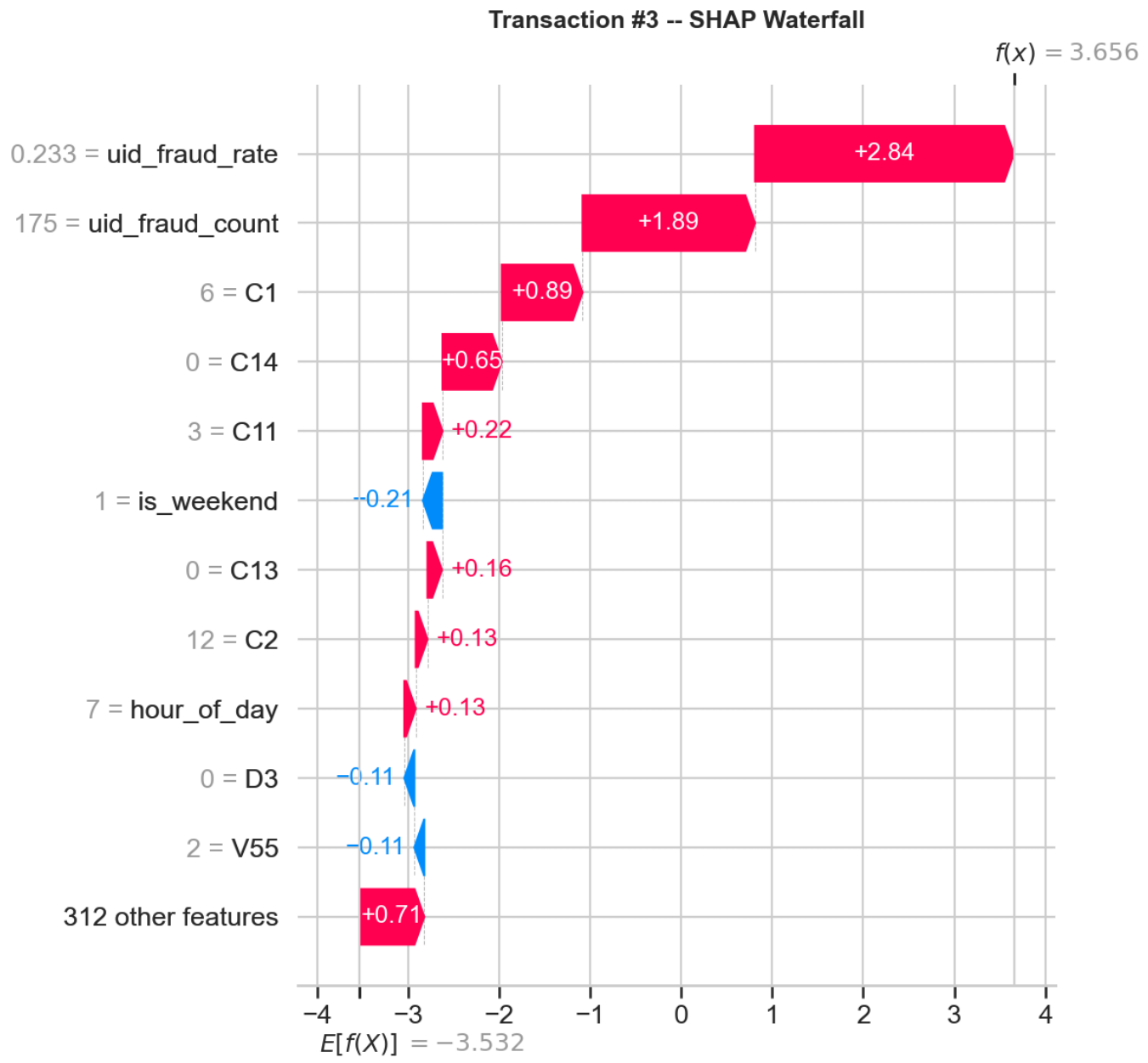
SHAP Waterfall -- Transaction #2

Individual feature contributions for a high-risk transaction:



SHAP Waterfall -- Transaction #3

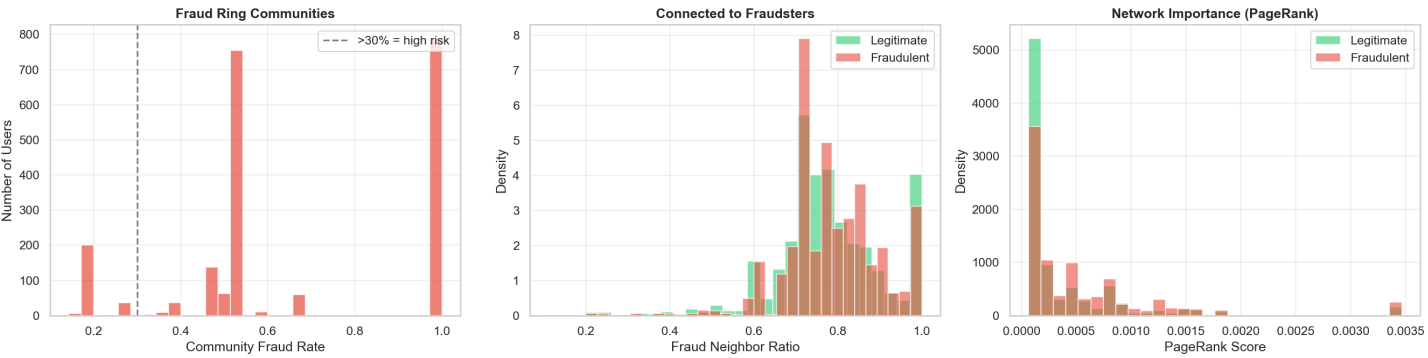
Individual feature contributions for a high-risk transaction:



Graph-Based Fraud Ring Analysis

Temporal graph analysis using NetworkX builds a relationship graph where nodes are users and edges connect users who share devices, addresses, or rare email domains. Louvain community detection identifies fraud rings. Features: PageRank, betweenness centrality, community fraud rate, fraud neighbor ratio.

FraudShield AI -- Graph-Based Fraud Ring Analysis



Adversarial Robustness Testing

FraudShield AI was tested against 5 realistic attack scenarios to ensure the model cannot be easily fooled by smart fraudsters.

Robustness Score: 100% (5/5 passed)

[PASS] amount_splitting: 71.1% -- Can model detect fraud when amounts are split into smaller transactions?

[PASS] time_evasion: 69.7% -- Can model detect fraud during business hours (time evasion)?

[PASS] device_spoofing: 72.2% -- Can model detect fraud from normal-looking devices (device spoofing)?

[PASS] threshold_sensitivity: 99.6% -- Is model performance stable across different decision thresholds?

[PASS] feature_perturbation: 96.6% -- Are predictions confident enough to resist small feature perturbations?

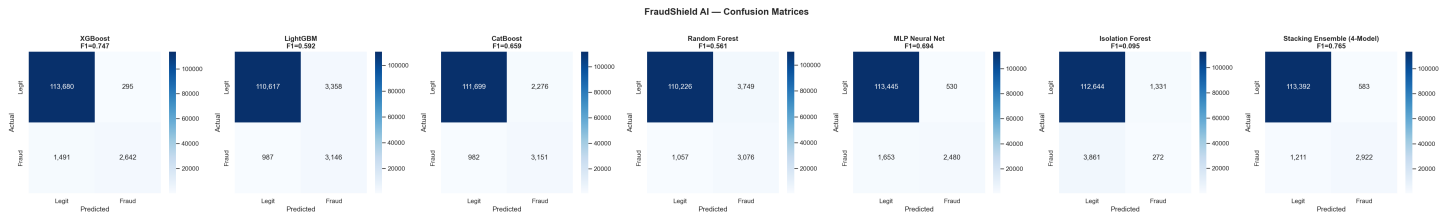
Business Impact -- ROI Analysis

Cost-benefit analysis for a mid-size bank (50M transactions/year):

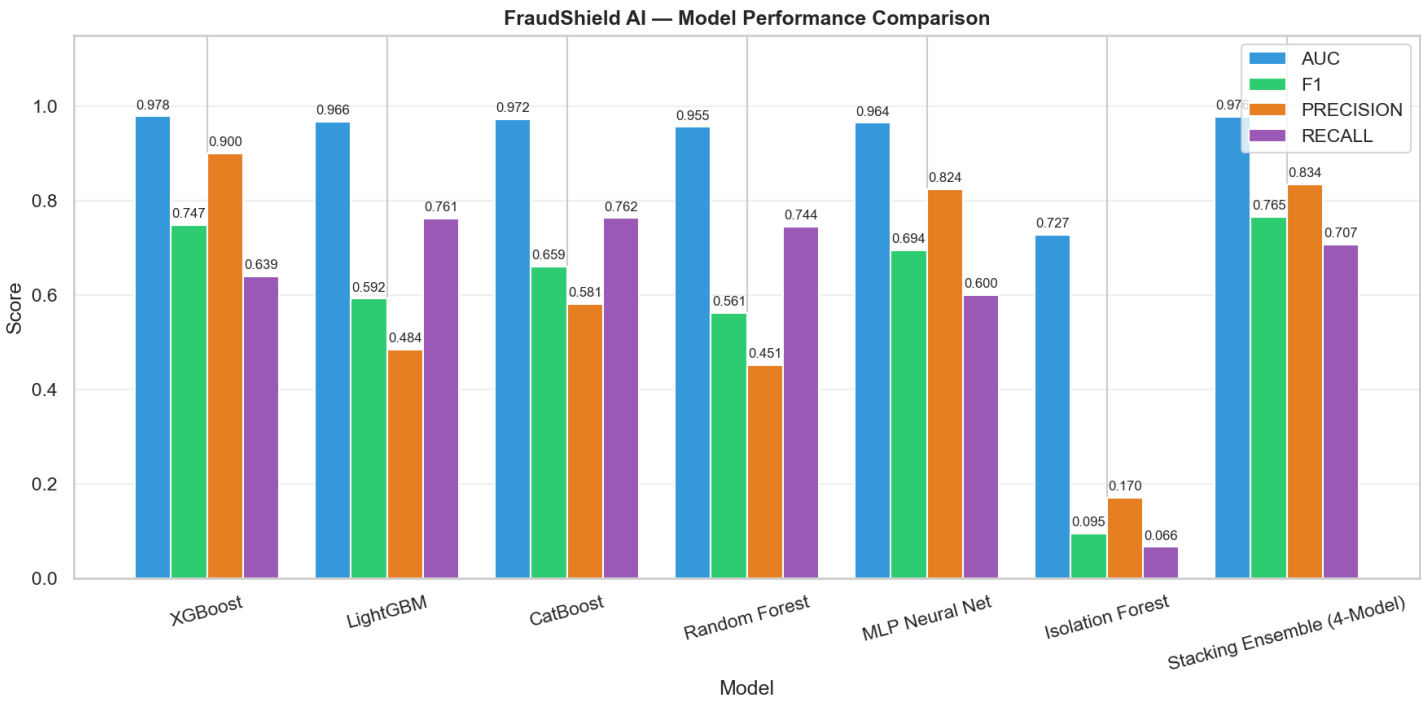
Annual Transactions:	50,000,000
Fraud Rate:	3.5%
Fraud Detection Rate:	70.7%
Precision:	83.4%
Frauds Prevented/Year:	1,237,237
Fraud Losses Avoided:	\$1,051,651,343
Baseline Annual Loss:	\$1,100,625,000
With FraudShield:	\$464,281,394

Annual Savings: \$636,343,606

Confusion Matrices



Metrics Comparison



Risk Score Distribution

